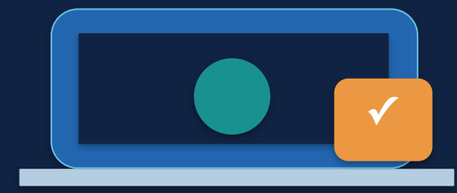
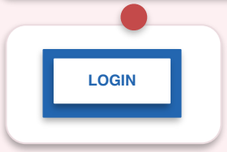
WEB SECURITY

Protecting websites, users and digital information



SECURE CODE · SAFER USERS · STRONGER WEB

COMMON WEB THREATS



Phishing

Fake messages or websites steal sensitive information.



SQL Injection

Malicious input manipulates database queries.



Cross-Site Scripting (XSS)

Unsafe scripts run in webpages viewed by users.



Weak Passwords

Easy-to-guess passwords expose user accounts.



Data Leaks

Private information becomes accessible to others.

HOW A WEB ATTACK HAPPENS

1. **Finds a weakness** →
2. **Exploits the flaw** →
3. **Accesses data / system** →
4. **Steals or changes data**

Example: phishing, SQL injection or XSS can exploit a weakness.

WAYS TO PROTECT WEBSITES



HTTPS

Encrypts data in transit



Strong passwords

Long, unique passwords



Input validation

Check data on client + server



Regular updates

Fix known vulnerabilities



Secure storage

Restrict access to sensitive data

SECURE DEVELOPMENT FLOW

1. Plan →
2. Develop
3. Test →
4. Deploy & Monitor

AUTHENTICATION vs AUTHORIZATION

AUTHENTICATION

Verifies who you are (e.g., password or MFA)

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AUTHORIZATION

Decides what you can do (e.g., access permissions)

WHAT I LEARNED

- Frontend checks help users; server-side validation is essential.
- Secure coding, HTTPS and updates reduce risk.
- Protecting data builds trust.

WHY IT MATTERS



Privacy
Protects data



Trust
Builds confidence



Access
Blocks misuse



Reliability
Safer services

KEY TAKEAWAY

A secure website is not just about technology — it protects people, privacy and trust.

Think before you click. Build with security in mind.

3CE2 B Batch
25DCE111. Dhanraj Solanki